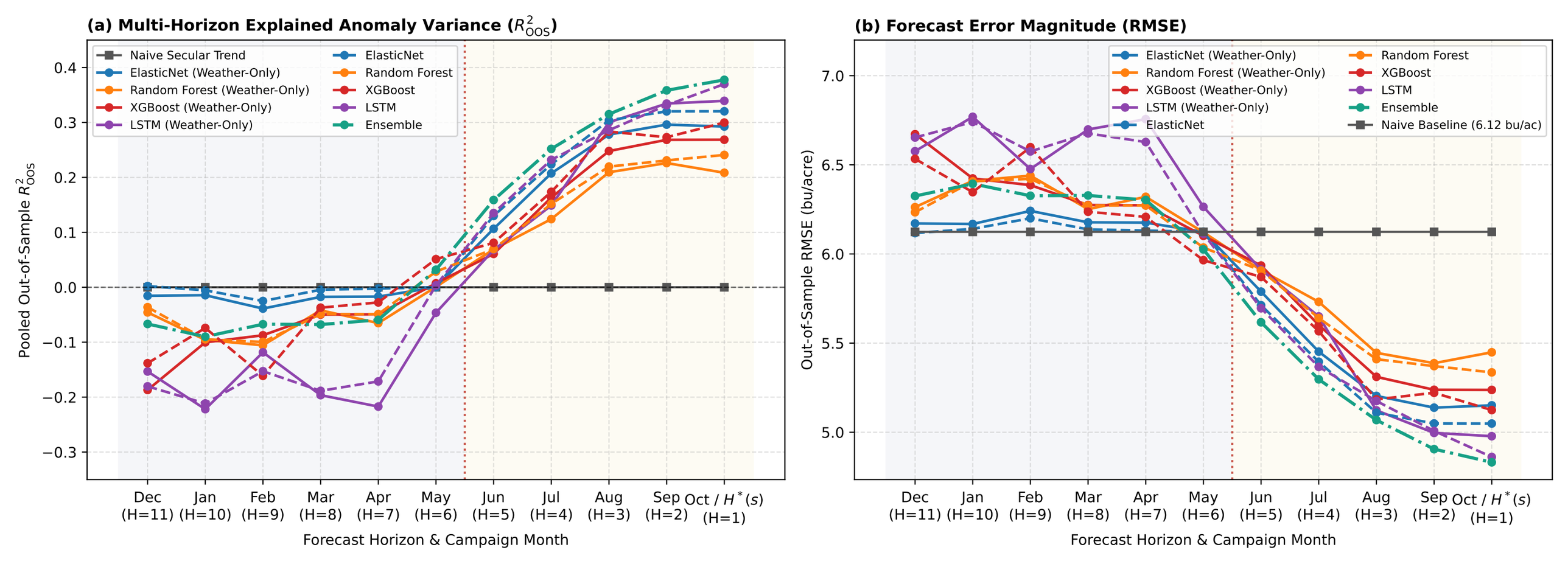
(a) Multi-Horizon Explained Anomaly Variance (R^2_{OOS})
Pooled Out-of-Sample R^2_{OOS}
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Naive Secular Trend
ElasticNet (Weather-Only)
Random Forest (Weather-Only)
XGBoost (Weather-Only)
LSTM (Weather-Only)
ElasticNet
Random Forest
XGBoost
LSTM
Ensemble
Dec (H=11)
Jan (H=10)
Feb (H=9)
Mar (H=8)
Apr (H=7)
May (H=6)
Jun (H=5)
Jul (H=4)
Aug (H=3)
Sep (H=2)
Oct / $H^*(s)$ (H=1)
Forecast Horizon & Campaign Month
(b) Forecast Error Magnitude (RMSE)
Out-of-Sample RMSE (bu/acre)
7.0
6.5
6.0
5.5
5.0
ElasticNet (Weather-Only)
Random Forest (Weather-Only)
XGBoost (Weather-Only)
LSTM (Weather-Only)
ElasticNet
Random Forest
XGBoost
LSTM
Ensemble
Naive Baseline (6.12 bu/ac)
Dec (H=11)
Jan (H=10)
Feb (H=9)
Mar (H=8)
Apr (H=7)
May (H=6)
Jun (H=5)
Jul (H=4)
Aug (H=3)
Sep (H=2)
Oct / $H^*(s)$ (H=1)
Forecast Horizon & Campaign Month